

SADAIVAL SINGH

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Education

Degree/Certificate	Institute/Board	CGPA/Percentage	Year
M. Eng Mechanical	McGill University	3.73/4	2025
B. Tech Major Mechanical, Minor ECE	Indian Institute of Technology, Guwahati	8.43/10	2021
Senior secondary	CBSE board	92.2%	2017
Secondary	CBSE board	9.4/10	2015

Experience

- Research Assistant — École de technologie supérieure Montréal***March 2026 – Present*
 - Project on improving the accuracy of Computed Laminography system.
- Research Associate — Polytechnique Montréal***June 2025 – Feb 2026*
 - Working with team on a novel elastic registration algorithm to accelerate CT image acquisition, reconstruction, and segmentation.
 - Built a Python-based UI for the backend API of the Elastic Registration system.
- Research Intern — Comet Group (Dragonfly Division)***May 2024 – Apr 2025*
 - Improved deep learning model accuracy for industrial vision applications and quantified measurement repeatability.
 - Automated code understanding and documentation by generating call graphs, type hints, and docstrings using LLM for RAG pipeline.
- Teaching Assistant — McGill University***Jan 2024 – Apr 2024*
 - Assisted in labs and grading for the course *MECH 383: Applied Electronics and Instrumentation*
- R&D Engineer — Ace Manufacturing Systems***Nov 2022 – Jun 2023*
 - Designed vision-based control systems for WAAM and PBF additive manufacturing.
 - Built a low-cost 3D scanner for part inspection using line-laser triangulation.
 - Developed a vision-driven tool measurement and breakage detection system in Python.
 - Optimized mechanism using constraint-based analysis and optimisation tools for WAAM machine.
- Assistant Manager / GTE / Intern — Bajaj Auto Ltd.***Jun 2020 – Oct 2022*
 - Assisted component development for engine castings, improving process efficiency and cost.
 - Designed tools, fixtures, and manufacturing processes ensuring quality and production targets.
 - Optimized paint shop equipment layout and reduced packaging costs through stress-based pallet redesign.
 - Analyzed critical wall thickness in blow-molded and rotomolded fuel tanks; designed universal handling fixtures.

Publications

- Singh, Sadaival, Ambrish Singh, Sajan Kapil, and Manas Das. "Utilization of a TSP solver for generating non-retractable, direction favoring toolpath for additive manufacturing." Additive Manufacturing (2022): 103126.
- Singh, Sadaival, Ambrish Singh, Sajan Kapil, and Manas Das. "Generation of Continuous and Sparse Space Filling Toolpath with Tailored density for Additive Manufacturing of Biomimetics." Computer-Aided Design (CAD) (2024): 103718

Projects

- Computer Vision in Additive Manufacturing** – IIT Guwahati, Prof. Sajan Kapil (Nov 2022 – Jun 2023): Implemented computer vision and deep learning algorithms for in-situ defect detection and melt-pool monitoring in WAAM and PBF processes.
- Morphing Wing Design Optimization** – McGill University (Jan 2024 – Apr 2024): Optimized aerodynamic, structural, and actuation subsystems of a morphing wing using XFOIL-coupled multi-objective optimization.

- **Acoustic Tactile Sensor Mapping** – *McGill University, Prof. Audrey Sedal (Sep 2024 – Apr 2025)*: Developed an acoustic-based tactile sensor and formulated the mapping problem as factor-graph SLAM using IMU and encoder data.
- **Area-Filling Method for 3D Printing** – *IIT Guwahati, Prof. Sajan Kapil (May 2019 – Present)*: Designed a TSP-based continuous toolpath algorithm and software for 2D geometry infill; results published at HI-AM 2021.
- **SAE Efficycle 2018** – *IIT Guwahati (Jun 2018 – Sep 2018)*: Built a three-wheeled hybrid vehicle with automatic electronic gear shifting; won Innovation Award (Rank 9/72).
- **Formula Student Vehicle** – *IIT Guwahati (Dec 2018 – Jan 2020)*: Led the design and manufacture of a Formula Bharat race car as team captain, overseeing suspension and electronics.
- **Hand Gesture Recognition** – *IIT Guwahati (Feb 2018)*: Implemented real-time gesture recognition using Python and OpenCV for camera-based interaction.

Technical skills

Programming Language	Python, C, C++, MATLAB
Miscellaneous	Machine Learning, Deep Learning, Computer Vision, ROS, PyTorch
CAD software	SolidWorks, Catia, NX, Solid Edge
Simulation software	Ansys, EDEM
Operating System	Windows, Linux

Key Courses Taken

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| • Statistical Computer Vision | • Probability, Statistics and Machine Learning |
| • Applied Robotics | • Fluid Mechanics |
| • Control Systems | • Machine Design |
| • Optimization methods in Engineering | • Soft computing in Engineering |
| • Industrial Engineering and Operations Research | • Kinematics of Machinery |

Leadership and Achievements

- **Team Captain, Formula Student IIT Guwahati (2018–2020)**: Led a 20-member team, managing design, manufacturing, and integration of a Formula Bharat race car; coordinated deadlines and ensured timely project delivery.
- **City Representative, Technothon 2018**: Organized Chandigarh exam center; secured 300+ student registrations and oversaw smooth event execution.
- **Best Innovation Award – SAE Efficycle 2018**: Designed and implemented an automatic electronic gear shifter, earning the prize money for Best Innovation Award and achieving 9th rank among 72 teams overall.
- **Group Mathematics Olympiad (2015)**: Selected among top 30 students in India for the Indian National Mathematics Olympiad (INMO).
- **Joint Entrance Examination (2017)**: Achieved All India Rank 2346 among 1.28 million candidates in JEE Advanced.